PERFECTNESS AND BASE-CHANGE STABILITY OF THE DERIVED PUSHFORWARD FOR FLAT PROPER FINITELY PRESENTED MORPHISMS

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Allowed standard prerequisites (editorial).

- The base-change tensor lemma computes the derived pushforward of a derived tensor as the derived tensor with the derived pushforward of the other factor, with perfectness and base-change stability. invoked at proof lines 7756-7757. Provides the machinery for both parts.

Source reference. Original lemma 0A1H: sources/stacks-perfect/pinned-original-source.tex, source lines 7614–7614. Cited at proof lines 7756-7757 for both parts; the two instantiations are spelled out inside the counted span.

Lemma 0.2. *Let S be a scheme. Let $f : X \to S$ be a proper morphism of finite presentation.*

(1) *Let $E \in D(\mathcal{O}_X)$ be perfect and f flat. Then Rf_*E is a perfect object of $D(\mathcal{O}_S)$ and its formation commutes with arbitrary base change.*
(2) *Let $\mathcal{G}$ be an $\mathcal{O}_X$-module of finite presentation, flat over S. Then $Rf_*\mathcal{G}$ is a perfect object of $D(\mathcal{O}_S)$ and its formation commutes with arbitrary base change.*

Proof. Special cases of Lemma 0A1H applied with (1) $\mathcal{G}^\bullet$ equal to $\mathcal{O}_X$ in degree 0 and (2) $E = \mathcal{O}_X$ and $\mathcal{G}^\bullet$ consisting of $\mathcal{G}$ sitting in degree 0. □

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